Gissel Dayana Pardo Montaguth

Postdoctoral Researcher in Extragalactic Astronomy

IAG – Universidade de São Paulo
Rua do Matão 1226, São Paulo, Brazil
•[regular] montaguth.g.c@gmail.com
ORCID: 0009-0003-1364-3590 Nationality: Colombian

Research interests

Galaxy evolution across environments (field, groups, compact groups, clusters and the cosmic web); morphological and structural transformation (bulges, discs and the mass–size relation); multi-band photometric surveys (S-PLUS, Rubin/LSST); star formation, nuclear activity and ionised gas in interacting galaxies.

Positions

2024–present **Postdoctoral Researcher – FAPESP Fellow**, *IAG, Universidade de São Paulo*, Brazil
FAPESP processes 2024/10923-3 and 2025/14602-0. Supervisor: Prof. C. Mendes de Oliveira.

2019–2020 **Young Researcher – Minciencias Talent Program**, *University of Antioquia*, Colombia
Interdisciplinary science communication for the Ophidism and Scorpionism Program.

Education

2020–2024 **Ph.D. in Astronomy**, *University of La Serena*, Chile
Supervisors: Drs. Antonela Monachesi and Sergio Torres-Flores. Thesis: how the environment affects galaxy evolution in dense environments such as compact groups compared with the field, from an integral multi-wavelength perspective.

2013–2019 **B.Sc. in Astronomy**, *University of Antioquia*, Colombia
Supervisor: Dr. Juan Carlos Muñoz-Cuartas. Thesis: ionisation of the ISM and star formation in minor mergers of galaxies.

Research visits

Jun 2026 **Institut d'Astrophysique de Paris** (host: G. Mamon) and **CEA Paris-Saclay, DAp** (host: E. Bertin), France. Invited seminars; discussions on integrating MorphoPLUS with SExtractor/SourceXtractor++.

2024 **Max Planck Institute for Astrophysics**, Garching, Germany (host: G. Kauffmann). Research internship: the impact of galaxy interactions in compact groups on their physical properties, especially ionised gas.

Collaborations

S-PLUS Southern Photometric Local Universe Survey: full member and active science contributor (galaxy evolution in groups and clusters, structural and morphological analysis with MorphoPLUS); involved in the SCALE project and the DR5 science verification.

Rubin/LSST Member of the LSST Galaxies Science Collaboration.

CHANCES CHilean Cluster galaxy Evolution Survey (4MOST): member of the morphology working group.

Publications

13 refereed papers (4 as first author; 1 in press) + 6 submitted; 94 citations, $h = 5$ (NASA ADS, Nov 2026). Author position given as (position/total). Full list on NASA ADS.

First-author refereed papers

2026 **Montaguth, G. P.** et al., *Galaxies caught in transition: the role of group environment in shaping the mass–size relation in the local Universe*, ApJ, 999, 160. doi:10.3847/1538-4357/ae42ca (1/11)

2026 **Montaguth, G. P.** et al., *Galaxy evolution in compact groups. III. Structural analysis of galaxies and dynamical state of non-isolated compact groups*, ApJ, 998, 91. doi:10.3847/1538-4357/ae2bdd (1/20)

2025 **Montaguth, G. P.** et al., *Galaxy evolution in compact groups. II. Witnessing the influence of major structures in their evolution*, A&A, 696, A240. doi:10.1051/0004-6361/202451198 (1/18)

2023 **Montaguth, G. P.** et al., *Galaxy evolution in compact groups – I. Revealing a transitional galaxy population through a multiwavelength approach*, MNRAS, 524, 5340. doi:10.1093/mnras/stad2235 (1/15)

Co-authored refereed papers

2026 Lima-Dias, C., et al. incl. **Montaguth, G. P.**, *Star formation and morphological trends in the Antlia cluster: probing environmental influence out to* $5R_{200}$, A&A, accepted (in press). arXiv:2609.23140 (3/51)

2026 Mendes de Oliveira, C., et al. incl. **Montaguth, G. P.**, *S-PLUS Clusters And Large-scale Environments (SCALE). I. A catalog of known clusters and groups in DR5 and a pilot study of Abell 4038*, ApJ, 1007, 202. doi:10.3847/1538-4357/ae84d0 (18/29)

2026 Vinicius Rodrigues de Lima, E., et al. incl. **Montaguth, G. P.**, *The S-PLUS Fifth Data Release: over 4500 square degrees of the southern sky and a multicolor view of the Hydra and Antlia galaxy clusters*, ApJ, 1007, 167. doi:10.3847/1538-4357/ae8d1f (15/59)

2026 Pallero, D., et al. incl. **Montaguth, G. P.**, *Characterising the formation and evolution of S0 galaxies (CaFES-0): revealing the origin of the mass–size relation of S0 galaxies*, A&A, 712, A123. doi:10.1051/0004-6361/202659852 (9/9)

2026 Dulcien, C., et al. incl. **Montaguth, G. P.**, *Caught in the web: galaxy mergers along cosmic filaments*, A&A, 708, L5. doi:10.1051/0004-6361/202558037 (27/34)

2026 Ortiz-Gómez, S., et al. incl. **Montaguth, G. P.**, *Star-forming compact groups: tracing the early evolutionary stages of compact group environments*, A&A, 707, A18. doi:10.1051/0004-6361/202556607 (4/16)

2024 Herpich, F. R., et al. incl. **Montaguth, G. P.**, *The Fourth S-PLUS Data Release: 12-filter photometry covering ∼3000 square degrees in the southern hemisphere*, A&A, 689, A249. doi:10.1051/0004-6361/202449725 (65/71)

2024 Haack, R. F., et al. incl. **Montaguth, G. P.**, *The S-PLUS Fornax Project (S+FP): SExtractor detection and measurement of nearby galaxies in large photometric surveys*, MNRAS, 530, 3195. doi:10.1093/mnras/stae1053 (11/14)

2024 Lima-Dias, C., et al. incl. **Montaguth, G. P.**, *Bulge-disc decomposition of the Hydra cluster galaxies in 12 bands*, MNRAS, 527, 5792. doi:10.1093/mnras/stad3571 (6/18)

Submitted

2026 **Montaguth, G. P.**, Mendes de Oliveira, C., et al., *Galaxies caught in transition II: structure and stellar populations respond differently to the group environment*, ApJ, under revision (referee report received) (1/14)

2026 Li, L., **Montaguth, G. P.**, et al., *S-PLUS Clusters And Large-scale Environments (SCALE): galaxy properties in a multi-cluster system at* $z \sim 0.04$ *out to* $5 \times R_{200}$, ApJ, submitted (2/17)

2026 Zamorano, S., **Montaguth, G. P.**, Torres-Flores, S., *Nuclear activity in resonance ringed galaxies: a higher AGN fraction triggered by secular evolution*, A&A, submitted (2/3)

2026 Thainá-Batista, J., et al. incl. **Montaguth, G. P.**, *S-PLUS Clusters and Large Scale Environments (SCALE): the interplay of mass, environment, and morphology in shaping star formation histories in the MKW4 cluster*, ApJ, submitted (3/16)

2026 da Silva, B. H., et al. incl. **Montaguth, G. P.**, *Interacting galaxies in MaNGA. I. Sample selection & Tully–Fisher relation*, MNRAS, submitted (7/9)

2026 Olave-Rojas, D. E., et al. incl. **Montaguth, G. P.**, *S-PLUS Cluster And Large-scale Environments (SCALE): IV. Environmental pre-processing and galaxy evolution around MKW6*, A&A, submitted (10/23)

Other

2020 Lopera Londoño, C., et al. incl. **Montaguth, G. P.**, *Arañas de Colombia: biología, envenenamiento y potenciales usos terapéuticos de su veneno* (interdisciplinary book). Online version.

Fellowships, grants and awards

Fellowships and grants

2026–2027 **CNPq – PIBIC** undergraduate research scholarship (USP project 2026-2363), awarded to the project I designed and supervise with V. K. Brasil Sanada.

2025 **FAPESP Postdoctoral Fellowship** (2025/14602-0), IAG/USP.

2024 **FAPESP Postdoctoral Fellowship** (2024/10923-3), IAG/USP.

2020 **ANID National Doctoral Scholarship** (Chile), to pursue my Ph.D. at the University of La Serena.

2019–2020 **Young Researchers and Innovators Scholarship**, Ministry of Science of Colombia (Minciencias) and University of Antioquia.

Observing time

2022B **Gemini South**, PI of programme GS-2022B-Q-210: *Witnessing galaxy transformation in compact groups: an IFU insight into the kinematic and physical properties of galaxies*.

Earlier awards

2016–2018 IAU scholarship for the International School for Young Astronomers (ISYA, 2018); Young Researchers Scholarship, University of Antioquia (2017); observational internship scholarship, Pico dos Dias Observatory, Brazil (2017; 7 nights); Dean's List, University of Antioquia (2016).

Supervision and mentoring

2026–2027 **V. K. Brasil Sanada** (undergraduate, IAG/USP; CNPq-PIBIC), supervisor. *The mass–size relation of galaxies in the cluster MCXC J1020.4–0631 out to $5R_{200}$: galactic structure and environmental pre-processing in the context of the SCALE project*.

2025–present **A. C. Saggioratto Osello** (undergraduate, Escola Politécnica / EESC, USP), supervisor. *Machine-learning-based validation of MorphoPLUS structural fitting in S-PLUS galaxy images*.

2025–present **L. Li** (Ph.D. student, IAG/USP; supervisor C. Mendes de Oliveira), research mentor (informal). Guided a thesis chapter on galaxies in clusters and their large-scale environments (SCALE), resulting in Li, Montaguth et al. (ApJ, submitted; 2nd author) and a joint analysis of Hydra and Antlia for S-PLUS DR5.

During Ph.D. **S. Ortiz-Gómez** (M.Sc. student, University of La Serena), research mentor (informal). Training in multi-band structural fitting with MorphoPLUS and GalfitM, resulting in Ortiz-Gómez et al. 2026, A&A, 707, A18 (4th author).

Ongoing **S. Zamorano** (student at the University of La Serena when the project started, in a course by S. Torres-Flores; now M.Sc. student), research mentor (informal). Guidance on AGN/ionisation diagnostics (BPT) and environment analysis of resonance-ringed galaxies, resulting in Zamorano, Montaguth & Torres-Flores (A&A, submitted; 2nd author).

Teaching

Jul 2026 Lecturer and instructor, workshop *From Image to Science: Exploring the Universe with Large Survey Data* (S-PLUS school for final-year undergraduates, IAG/USP): lecture on Source Extractor and building catalogues from astronomical images; hands-on sessions on astronomical databases and catalogue building.

2018 Teaching assistant, astronomy basics course for non-science students, University of Antioquia.

Talks

Invited seminars

Jun 2026 *Caught in the act: a multi-wavelength view of galaxy transformation in groups*, Institut d'Astrophysique de Paris, France.

Jun 2026 *Galaxy evolution in groups: multi-wavelength studies of morphology and star formation*, CEA Paris-Saclay (DAp), France.

Dec 2024 *Galaxy evolution in compact groups*, FCAG Seminar Series, Universidad Nacional de La Plata, Argentina.

Selected contributed talks

Nov 2026 Poster: *Structural transformation of galaxies in dense environments: from S-PLUS to Rubin LSST*, 3rd LSST@LATAM Meeting.

Oct 2026 *Caught in the act: a multi-wavelength view of galaxy transformation in groups*, Annual Meeting of the Brazilian Astronomical Society (SAB).

Oct 2026 *Unveiling star formation histories in the MKW4 cluster using S-PLUS data* (on behalf of Thainá-Batista et al.), Annual Meeting of the Brazilian Astronomical Society (SAB).

Aug 2026 *Environmental imprints on galaxy structure and star formation in groups*, 21st S-PLUS Collaboration Meeting.

Jun 2026 *Galaxy evolution in groups: multi-wavelength studies of morphology and star formation*, LSST Galaxies Science Collaboration Meeting, Paris, France.

Aug 2025 *Galaxies under pressure: the role of galaxy groups in shaping their mass–size relation*, 20th S-PLUS Collaboration Meeting.

Jun 2024 *Galaxy evolution in compact groups: revealing the peculiar population and major structural influences*, LSST@LATAM meeting, La Serena, Chile.

Sep 2023 *Exploring the impact of the environment where CGs are located on the star formation of their galaxies*, 3rd Australia–ESO meeting *Galaxy Transformation Across Space and Time*, Canberra, Australia.

May 2023 *A peculiar galaxy population: morphological transformation in compact groups*, Galaxy Groups and Clusters III, La Serena, Chile.

Nov 2022 *The role of dense environments in the morphological transformation of galaxies: compact groups in Stripe 82*, S-PLUS meeting, Rio de Janeiro, Brazil.

+ 9 additional talks and posters (2018–2023) at S-PLUS meetings, SOCHIAS, CATA, CoCoA, ISYA and other venues.

Public outreach

2018–2025 Radio host, programme *El Laboratorio*, educational radio system of the University of Antioquia.

Jan 2026 Planetarium of Cali: *Ciudades en el Universo: grupos y cúmulos de galaxias* (Noche de Ciencia).

Apr 2022 Planetarium of Medellín: *Galaxias: historias de vecindarios espaciales* (Ciencia en Bicicleta).

Jul 2020 Planetarium of Medellín, *El Cielo Esta Noche*: Hablemos de cometas.

2020 Two videos (Moon phases; Saturn–Moon conjunction) for the Conociendo el Cielo project, University of La Serena.

2020 Podcaster, as a member of the science-communication group *Quimikomedia*: La Verdad Absoluta del Peluche.

Service

2024 External referee for the Canadian Time Allocation Committee (CanTAC), CFHT observing proposals, semester 2025A.

2023 Local Organizing Committee, *Mock La Serena Meeting*, University of La Serena (16–19 October).

2023 Local Organizing Committee, *Galaxy Groups and Clusters III*, University of La Serena (15–19 May).

Skills

Own software **MorphoPLUS**: automated multi-band parametric fitting of galaxy light profiles (bash, Python, GalfitM) across the 12 S-PLUS filters.

Analysis Multi-band structural fitting and bulge–disc decomposition; galaxy dynamics; emission-line diagnostics (BPT), electron density and SFR estimation; statistical analysis in Python; large survey databases (S-PLUS, SDSS, Legacy Survey, Rubin/LSST).

Observations Photometric and spectroscopic reduction (IRAF); spectral extraction and calibration; observing proposals (Gemini PI); 7 nights at the Pico dos Dias Observatory.

Programming Python, bash, LaTeX; Galfit, GalfitM, SExtractor.

Languages Spanish (native), Portuguese (fluent), English (full professional proficiency).

References

Available upon request.